



Magruder Fertilizer Proficiency Testing Program – Striving for Analytical Excellence

Proficiency Testing Program – Data Reporting – Analytes/Methods

Analytes/Methods List

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For Sample #230311:NPK

in
2023 Scheme R

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Total: 262

Method Code	Analyte Name	Method Description	Unit	References	Active
001.10	Ammoniacal N	Magnesium Oxide Method	%	920.03	12/15/2014
001.99	Ammoniacal N	Other	%		12/15/2014
002.10	Nitrate N	Robertson	%	930.01	12/15/2014
002.20	Nitrate N	Jones Modified	%	930.02	12/15/2014
002.99	Nitrate N	Other	%		12/15/2014
003.10	Water Insoluble N	Method I	%	945.01	12/15/2014
003.20	Water Insoluble N	Method II (Katz)	%	970.04	12/15/2014
003.99	Water Insoluble N	Other	%		12/15/2014
004.00	N Activity Index	Urea-Formaldehyde Compounds	%	955.05	12/15/2014
005.00	Urea N	Urease (as N)	%	959.03	12/15/2014
005.10	Urea N	HPLC (as N), H2O mobile phase	%	AOAC 983.01	12/17/2019
005.20	Urea N	HPLC (as N), 85% acetonitrile mobile phase	%	AOAC 2003.14 / ISO 19746	12/17/2019
005.99	Urea N	Other	%		12/15/2014
006.00	Biuret N	AA (as N)	%	976.01	12/15/2014
006.10	Biuret N	Spectrophotometric (as N)	%	960.04	12/15/2014
006.99	Biuret N	Other	%		12/15/2014
007.00	Urea	Urease (as Urea)	%	959.03	12/15/2014
007.10	Urea	HPLC, H2O mobile phase	%	AOAC 983.01	12/17/2019
007.20	Urea	HPLC, 85% acetonitrile mobile phase	%	AOAC 2003.14 / ISO 19746	12/17/2019
007.99	Urea	Other	%		12/15/2014
008.00	Biuret	AA (as Biuret)	%	976.01	12/15/2014
008.10	Biuret	Spectrophotometric (as Biuret)	%	960.04	12/15/2014
008.99	Biuret	Other	%		12/15/2014
009.10	Ammoniacal Plus Nitrate N	Devarda	%	892.01	12/15/2014
009.99	Ammoniacal Plus	Other	%		12/15/2014

009.99	Nitrate N	Other	%		12/15/2014
010.10	Total N	Reduced Iron	%		12/15/2014
010.11	Total N	Modified Comprehensive	%	978.02	12/15/2014
010.12	Total N	Salicylic	%	955.04D	12/15/2014
010.16	Total N	Raney	%	970.03	12/15/2014
010.17	Total N	Comprehensive	%	970.02	12/15/2014
010.60	Total N	Combustion	%	993.13	12/15/2014
010.99	Total N	Other	%		12/15/2014
020.10	Total P2O5	Gravimetric Quinolinium MolybdoP	%	962.02	12/15/2014
020.20	Total P2O5	Spectrophotometric MolybdovanadoP	%	958.01	12/15/2014
020.30	Total P2O5	Alkalimetric Quinolinium MolybdoP	%	969.02	12/15/2014
020.40	Total P2O5	Automated	%	978.01	12/15/2014
020.50	Total P2O5	ICP	%		12/15/2014
020.60	Total P2O5	AFPC No.3B	%		12/15/2014
020.99	Total P2O5	Other	%		12/15/2014
030.10	Citrate Insoluble P2O5	Gravimetric Quinolinium MolybdoP	%	963.03	12/15/2014
030.20	Citrate Insoluble P2O5	Spectrophotometric MolybdovanadoP	%	963.03	12/15/2014
030.30	Citrate Insoluble P2O5	Alkalimetric Quinolinium MolybdoP	%	963.03	12/15/2014
030.40	Citrate Insoluble P2O5	Automated, test portion 963.03 A-B	%	978.01	12/15/2014
030.50	Citrate Insoluble P2O5	ICP, test portion 963.03 A-B	%		12/15/2014
030.99	Citrate Insoluble P2O5	Other	%		12/15/2014
040.10	Indirect Available P2O5	Gravimetric Quinolinium	%	960.02	12/15/2014
040.20	Indirect Available P2O5	Spectrophotometric	%	960.02	12/15/2014
040.30	Indirect Available P2O5	Alkalimetric Quinolinium MolybdoP	%	960.02	12/15/2014
040.40	Indirect Available P2O5	Automated	%	960.02	12/15/2014
040.50	Indirect Available P2O5	ICP	%		12/15/2014
040.99	Indirect Available P2O5	Other	%		12/15/2014
041.10	Direct Available P2O5	Gravimetric Quinolinium	%	960.03	12/15/2014
041.11	Direct Available P2O5	Gravimetric Quinolinium, Citrate-EDTA Ext.	%	993.31	2/20/2015
	Direct Available				

041.20	Direct Available P2O5	Spectrophotometric	%	960.03	12/15/2014
041.21	Direct Available P2O5	Spectrophotometric, Citrate-EDTA Ext.	%	993.31	2/20/2015
041.30	Direct Available P2O5	Alkalimetric Quinolinium MolybdoP	%	960.03	12/15/2014
041.31	Direct Available P2O5	Alkalimetric Quinolinium MolybdoP, Citrate-EDTA Ext.	%		2/20/2015
041.40	Direct Available P2O5	Automated	%	978.01	8/19/2019
041.50	Direct Available P2O5	ICP	%		12/15/2014
041.51	Direct Available P2O5	ICP, Citrate-EDTA Ext.	%	2015.18	2/20/2015
041.99	Direct Available P2O5	Other	%		12/15/2014
042.10	Water Soluble OrthoP As P2O5	Spectrophotometric	%	970.01	12/15/2014
042.99	Water Soluble OrthoP As P2O5	Other	%		12/15/2014
043.99	Polyphosphate	Other	%		12/15/2014
048.10	Water Soluble P2O5	Gravimetric Quinolinium	%	962.03	12/15/2014
048.20	Water Soluble P2O5	Spectrophotometric	%	970.01	12/15/2014
048.30	Water Soluble P2O5	Alkalimetric Quinolinium MolybdoP	%	962.04	12/15/2014
048.40	Water Soluble P2O5	Automated	%		2/20/2015
048.50	Water Soluble P2O5	ICP	%		12/15/2014
048.99	Water Soluble P2O5	Other	%		12/15/2014
050.00	Soluble K2O	STPB Oxalate	%	958.02	12/15/2014
050.10	Soluble K2O	STPB Citrate	%	969.04	12/15/2014
050.30	Soluble K2O	AA (Oxalate)	%		12/15/2014
050.31	Soluble K2O	AA (Citrate)	%		12/15/2014
050.32	Soluble K2O	AA (Citrate-EDTA)	%		12/15/2014
050.50	Soluble K2O	ICP (Oxalate)	%		12/15/2014
050.51	Soluble K2O	ICP (Citrate)	%		12/15/2014
050.52	Soluble K2O	ICP (Citrate-EDTA)	%	2015.18	12/15/2014
050.60	Soluble K2O	Flame Photometric (Oxalate)	%	983.02	12/15/2014
050.61	Soluble K2O	Flame Photometric (Citrate)	%	983.02	12/15/2014
050.62	Soluble K2O	Flame Photometric (Citrate-EDTA)	%	983.02	12/15/2014
050.99	Soluble K2O	Other	%		12/15/2014
060.00	Water (Free)	Vacuum Oven	%	965.08B	12/15/2014
060.10	Water (Free)	Vacuum Desiccation	%	965.08A	12/15/2014
060.20	Water (Free)	Karl Fischer	%	972.01	12/15/2014
060.30	Water (Free)	AFPC No. 2B (105°C oven for 2 hours. 5a sample)	%		12/15/2014

060.99	Water (Free)	Other	%		12/15/2014
101.00	Acid Soluble Ca	AA, inorganic 965.09	%	965.09	12/15/2014
101.01	Acid Soluble Ca	AA, organic 965.09	%	965.09	12/15/2014
101.02	Acid Soluble Ca	AA, 965.09, test portion 2006.03A-C	%		12/15/2014
101.03	Acid Soluble Ca	AA, 965.09, test portion 2006.03A-C w/dual acid	%		12/15/2014
101.30	Acid Soluble Ca	ICP, test portion inorganic 965.09	%		12/15/2014
101.31	Acid Soluble Ca	ICP, test portion organic 965.09	%		12/15/2014
101.32	Acid Soluble Ca	ICP, test portion 2006.03A-C	%		12/15/2014
101.33	Acid Soluble Ca	ICP, 2017.02	%	2017.02	12/15/2014
101.70	Acid Soluble Ca	Titrimetric	%	945.03	12/15/2014
101.99	Acid Soluble Ca	Other	%		12/15/2014
121.00	Acid Soluble Mg	AA, inorganic 965.09	%	965.09	12/15/2014
121.01	Acid Soluble Mg	AA, organic 965.09	%	965.09	12/15/2014
121.02	Acid Soluble Mg	AA, 965.09, test portion 2006.03A-C	%		12/15/2014
121.03	Acid Soluble Mg	AA, 965.09, test portion 2006.03A-C w/dual acid	%		12/15/2014
121.30	Acid Soluble Mg	ICP, test portion inorganic 965.09	%		12/15/2014
121.31	Acid Soluble Mg	ICP, test portion organic 965.09	%		12/15/2014
121.32	Acid Soluble Mg	ICP, test portion 2006.03A-C	%		12/15/2014
121.33	Acid Soluble Mg	ICP, 2017.02	%	2017.02	12/15/2014
121.70	Acid Soluble Mg	Titrimetric (EDTA)	%	964.01	12/15/2014
121.99	Acid Soluble Mg	Other	%		12/15/2014
131.00	Water Soluble Mg	AA	%	965.09	12/15/2014
131.30	Water Soluble Mg	ICP	%		12/15/2014
131.70	Water Soluble Mg	Titrimetric (EDTA)	%	937.02	12/15/2014
131.99	Water Soluble Mg	Other	%		12/15/2014
143.00	Elemental S	Gravimetric Sulfur - carbon disulfide soluble sulfur	%		12/15/2014
143.99	Elemental S	Other	%		12/15/2014
145.00	Sulfate S, HCl soluble	Gravimetric Sulfur - sulfate form	%		12/15/2014
145.99	Sulfate S, HCl soluble	Other	%		12/15/2014
146.00	Total S in Liquid	Gravimetric - sulfate, sulfite, thiosulfate, and elemental	%	980.02b	12/15/2014
146.99	Total S in Liquid	Other	%		12/15/2014
147.00	Total S in Urea/Formulations	Gravimetric	%	980.02c	12/15/2014

147.99	Total S in Urea/Formulations	Other (Identify)	%		12/15/2014
148.00	Total S	Combustion, 2017.08	%	2017.08	12/15/2014
148.01	Total S	Gravimetric - sulfate and elemental	%	980.02a	2/20/2015
148.02	Total S	Turbidimetric, w/Br digestion, modification of JAOAC 63.845	%	JAOAC 63.845	12/15/2014
148.03	Total S	Spectrometric, w/Br digestion	%		12/15/2014
148.04	Total S	ICP, w/Br digestion	%		12/15/2014
148.05	Total S	Thermotitration, w/Br digestion	%		12/15/2014
148.06	Total S	Ion Exchange, w/Br digestion	%		12/15/2014
148.07	Total S	ICP, test portion as in 2017.02	%		1/23/2015
148.99	Total S	Other	%		12/15/2014
149.02	S - HNO3 soluble	Turbidimetric, modification of JAOAC 63.845	%		12/15/2014
149.03	S - HNO3 soluble	Spectrometric	%		12/15/2014
149.04	S - HNO3 soluble	ICP	%		12/15/2014
149.05	S - HNO3 soluble	Thermotitration	%		12/15/2014
149.06	S - HNO3 soluble	Ion Exchange	%		12/15/2014
149.99	S - HNO3 soluble	Other	%		12/15/2014
151.00	Acid Soluble As	AA, test portion 2006.03 modified w/dual acid	ppm		12/15/2014
151.30	Acid Soluble As	ICP	ppm		2/20/2015
151.32	Acid Soluble As	ICP, 2006.03	ppm	2006.03	12/15/2014
151.33	Acid Soluble As	ICP, 2017.02	ppm	2017.02	12/15/2014
151.34	Acid Soluble As	ICP, EPA 3050B/6010C	ppm		2/20/2015
151.99	Acid Soluble As	Other	ppm		12/15/2014
165.00	Acid Soluble B	Spectrophotometric	%	982.01	12/15/2014
165.30	Acid Soluble B	ICP, test portion in 982.01	%		2/20/2015
165.70	Acid Soluble B	Titrimetric	%	949.02	12/15/2014
165.99	Acid Soluble B	Other	%		12/15/2014
171.00	Water Soluble B	AA, JAOAC 52.950	%	JAOAC 52.590	12/15/2014
171.10	Water Soluble B	Spectrophotometric	%	982.01	12/15/2014
171.70	Water Soluble B	Titrimetric	%	949.03	12/15/2014
171.99	Water Soluble B	Other	%		12/15/2014
181.00	Acid Soluble Cd	AA, test portion 2006.03 modified w/dual acid	ppm		12/15/2014
181.30	Acid Soluble Cd	ICP	ppm		2/20/2015
181.32	Acid Soluble Cd	ICP, 2006.03	ppm	2006.03	12/15/2014
181.33	Acid Soluble Cd	ICP, 2017.02	ppm	2017.02	12/15/2014
181.34	Acid Soluble Cd	ICP, EPA 3050B/6010C	ppm		2/20/2015
181.99	Acid Soluble Cd	Other	ppm		12/15/2014
190.00	Water Soluble Cl	Titrimetric	%	928.02	12/15/2014

190.99	Water Soluble Cl	Other	%		12/15/2014
191.00	Acid Soluble Cr	AA, test portion 2006.03 modified w/dual acid	ppm		12/15/2014
191.30	Acid Soluble Cr	ICP	ppm		2/20/2015
191.32	Acid Soluble Cr	ICP, 2006.03	ppm	2006.03	12/15/2014
191.33	Acid Soluble Cr	ICP, 2017.02	ppm	2017.02	12/15/2014
191.34	Acid Soluble Cr	ICP, EPA 3050B/6010C	ppm		2/20/2015
191.99	Acid Soluble Cr	Other	ppm		12/15/2014
202.00	Acid Soluble Co	AA, test portion 2006.03 modified w/dual acid	ppm		12/15/2014
202.20	Acid Soluble Co	Colorimetric	ppm	965.11	12/15/2014
202.30	Acid Soluble Co	ICP	ppm		2/14/2015
202.32	Acid Soluble Co	ICP, 2006.03	ppm	2006.03	12/15/2014
202.33	Acid Soluble Co	ICP, 2017.02	ppm	2017.02	12/15/2014
202.34	Acid Soluble Co	ICP, EPA 3050B/6010C	ppm		2/20/2015
202.99	Acid Soluble Co	Other	ppm		12/15/2014
221.00	Acid Soluble Cu	AA, inorganic 965.09	%	965.09	12/15/2014
221.01	Acid Soluble Cu	AA, organic 965.09	%	965.09	2/20/2015
221.02	Acid Soluble Cu	AA, 965.09, test portion 2006.03A-C	%		2/20/2015
221.03	Acid Soluble Cu	AA, 965.09, test portion 2006.03A-C w/dual acid	%		2/20/2015
221.10	Acid Soluble Cu	Titrimetric	%	942.01	12/15/2014
221.30	Acid Soluble Cu	ICP, test portion inorganic 965.09	%		12/15/2014
221.31	Acid Soluble Cu	ICP, test portion organic 965.09	%		2/20/2015
221.32	Acid Soluble Cu	ICP, test portion 2006.03A-C	%		12/15/2014
221.33	Acid Soluble Cu	ICP, 2017.02	%	2017.02	12/15/2014
221.99	Acid Soluble Cu	Other	%		12/15/2014
241.00	Acid Soluble Fe	AA, inorganic 965.09	%	965.09	12/15/2014
241.01	Acid Soluble Fe	AA, organic 965.09	%	965.09	2/20/2015
241.02	Acid Soluble Fe	AA, 965.09, test portion 2006.03A-C	%		2/20/2015
241.03	Acid Soluble Fe	AA, 965.09, test portion 2006.03A-C w/dual acid	%		2/20/2015
241.20	Acid Soluble Fe	Colorimetric	%		12/15/2014
241.30	Acid Soluble Fe	ICP, test portion inorganic 965.09	%		12/15/2014
241.31	Acid Soluble Fe	ICP, test portion organic 965.09	%		2/20/2015
241.32	Acid Soluble Fe	ICP, test portion 2006.03A-C	%		2/20/2015
241.33	Acid Soluble Fe	ICP, 2017.02	%	2017.02	2/20/2015
241.70	Acid Soluble Fe	Titrimetric	%	967.01	12/15/2014

241.99	Acid Soluble Fe	Other	%		12/15/2014
251.00	Acid Soluble Pb	AA, test portion 2006.03 modified w/dual acid	ppm		12/15/2014
251.30	Acid Soluble Pb	ICP	ppm		12/15/2014
251.32	Acid Soluble Pb	ICP, 2006.03	ppm	2006.03	2/20/2015
251.33	Acid Soluble Pb	ICP, 2017.02	ppm	2017.02	2/20/2015
251.34	Acid Soluble Pb	ICP, EPA 3050B/6010C	ppm		2/20/2015
251.99	Acid Soluble Pb	Other	ppm		12/15/2014
261.00	Acid Soluble Mn	AA, Mn2+ only	%	972.02a	12/15/2014
261.11	Acid Soluble Mn	AA, total Mn	%	972.02b	12/15/2014
261.12	Acid Soluble Mn	AA, inorganic 965.09	%	965.09	2/20/2015
261.13	Acid Soluble Mn	AA, organic 965.09	%	965.09	2/20/2015
261.14	Acid Soluble Mn	AA, 965.09, test portion 2006.03A-C	%		2/20/2015
261.15	Acid Soluble Mn	AA, 965.09, test portion 2006.03A-C w/dual acid	%		2/20/2015
261.30	Acid Soluble Mn	ICP, test portion 972.02a	%		12/15/2014
261.31	Acid Soluble Mn	ICP, test portion 972.02b	%		12/15/2014
261.32	Acid Soluble Mn	ICP, test portion inorganic 965.09	%		2/20/2015
261.33	Acid Soluble Mn	ICP, test portion organic 965.09	%		2/20/2015
261.34	Acid Soluble Mn	ICP, test portion 2006.03A-C	%		2/20/2015
261.35	Acid Soluble Mn	ICP, 2017.02	%	2017.02	2/20/2015
261.40	Acid Soluble Mn	Bismuthate	%		12/15/2014
261.50	Acid Soluble Mn	Spectrometric	%	940.02	12/15/2014
261.99	Acid Soluble Mn	Other	%		12/15/2014
271.00	Water Soluble Mn	AA	%	972.03	12/15/2014
271.30	Water Soluble Mn	ICP, Ext. 972.03	%		12/15/2014
271.99	Water Soluble Mn	Other	%		12/15/2014
281.00	Acid Soluble Hg	AA	ppm		12/15/2014
281.30	Acid Soluble Hg	ICP	ppm		12/15/2014
281.99	Acid Soluble Hg	Other	ppm		12/15/2014
289.00	Acid Soluble Mo	AA, test portion 2006.03 modified w/dual acid	ppm		12/15/2014
289.30	Acid Soluble Mo	ICP	ppm		12/15/2014
289.32	Acid Soluble Mo	ICP, 2006.03	ppm	2006.03	2/20/2015
289.33	Acid Soluble Mo	ICP, 2017.02	ppm	2017.02	2/20/2015
289.34	Acid Soluble Mo	ICP, EPA 3050B/6010C	ppm		2/20/2015
289.99	Acid Soluble Mo	Other	ppm		12/15/2014
291.00	Acid Soluble Ni	AA, test portion 2006.03 modified w/dual acid	ppm		12/15/2014
291.30	Acid Soluble Ni	ICP	ppm		12/15/2014

291.32	Acid Soluble Ni	ICP, 2006.03	ppm	2006.03	2/20/2015
291.33	Acid Soluble Ni	ICP, 2017.02	ppm	2017.02	2/20/2015
291.34	Acid Soluble Ni	ICP, EPA 3050B/6010C	ppm		2/20/2015
291.99	Acid Soluble Ni	Other	ppm		12/15/2014
301.00	Acid Soluble Se	AA, test portion 2006.03 modified w/dual acid	ppm		12/15/2014
301.30	Acid Soluble Se	ICP	ppm		12/15/2014
301.32	Acid Soluble Se	ICP, 2006.03	ppm	2006.03	2/20/2015
301.33	Acid Soluble Se	ICP, 2017.02	ppm	2017.02	2/20/2015
301.34	Acid Soluble Se	ICP, EPA 3050B/6010C	ppm		2/20/2015
301.99	Acid Soluble Se	Other	ppm		12/15/2014
311.00	Sodium	AA	%	983.04	12/15/2014
311.30	Sodium	Flame Photometric	%	974.01	12/15/2014
311.32	Sodium	ICP, 2006.03 test portion	%		2/20/2015
311.33	Sodium	ICP, test portion as in 2017.02	%		2/20/2015
311.99	Sodium	Other	%		12/15/2014
321.00	Acid Soluble Zn	AA, inorganic 965.09	%	965.09	12/15/2014
321.01	Acid Soluble Zn	AA, organic 965.09	%	965.09	2/20/2015
321.02	Acid Soluble Zn	AA, 965.09, test portion 2006.03A-C	%		2/20/2015
321.03	Acid Soluble Zn	AA, 965.09, test portion 2006.03A-C w/dual acid	%		2/20/2015
321.10	Acid Soluble Zn	Gravimetric	%	942.02	12/15/2014
321.20	Acid Soluble Zn	Zincon Ion Exchange	%	973.01	12/15/2014
321.30	Acid Soluble Zn	ICP, test portion inorganic 965.09	%		12/15/2014
321.31	Acid Soluble Zn	ICP, test portion organic 965.09	%		2/20/2015
321.32	Acid Soluble Zn	ICP, test portion 2006.03A-C	%		2/20/2015
321.33	Acid Soluble Zn	ICP, 2017.02	%	2017.02	2/20/2015
321.99	Acid Soluble Zn	Other	%		12/15/2014
325.00	Water Soluble Zn	AA	%		12/15/2014
325.10	Water Soluble Zn	Gravimetric	%		12/15/2014
325.30	Water Soluble Zn	ICP	%		12/15/2014
325.99	Water Soluble Zn	Other	%		12/15/2014
431.10	Soluble Si in nonliquids	Na2CO3-NH4NO3 extraction, spectrophometry	%	JAOAC 96(2):251-259	9/16/2019
431.99	Soluble Si in nonliquids	Other	%		9/16/2019
441.00	Humic Acids	Gravimetry, Ash-free material dissolved in alkali and precipitated in acid	%	ISO 19822:2018	12/17/2019
441.10	Humic Acids	Gravimetry, Dissolved in alkali and precipitated in acid	%	California Dept. of Agric.	8/25/2020

441.99	Humic Acids	Other	%	ISO 19822:2018	12/17/2019
451.00	Hydrophobic Fulvic Acids	Gravimetry, Material dissolved in acid that bind to hydrophobic resin	%		12/17/2019
451.99	Hydrophobic Fulvic Acids	Other	%		12/17/2019